1 Find Sum of Array Elements

```
public class SumArray {
   public static void main(String[] args) {
      int[] arr = {10, 20, 30, 40};
      int sum = 0;

      for (int i = 0; i < arr.length; i++) {
         sum += arr[i];
      }

      System.out.println("Sum = " + sum);
   }
}
```

2. Find Largest Element

```
public class LargestElement {
   public static void main(String[] args) {
      int[] arr = {12, 45, 7, 89, 23};
      int max = arr[0];

      for (int i = 1; i < arr.length; i++) {
         if (arr[i] > max) {
            max = arr[i];
         }
      }
```

```
        System.out.println("Largest = " + max);
    }
}
```

3. Find Smallest Element

```
public class SmallestElement {
    public static void main(String[] args) {
        int[] arr = {12, 45, 7, 89, 23};
        int min = arr[0];

        for (int i = 1; i < arr.length; i++) {
            if (arr[i] < min) {
                min = arr[i];
            }
        }

        System.out.println("Smallest = " + min);
    }
}
```

4. Reverse an Array

```
public class ReverseArray {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5};

        for (int i = arr.length - 1; i >= 0; i--) {
```

```
            System.out.print(arr[i] + " ");
        }
    }
}
```

5. Search an Element

```
public class SearchElement {
    public static void main(String[] args) {
        int[] arr = {10, 20, 30, 40};
        int key = 30;
        boolean found = false;

        for (int i = 0; i < arr.length; i++) {
            if (arr[i] == key) {
                found = true;
                break;
            }
        }

        if (found)
            System.out.println("Element Found");
        else
            System.out.println("Element Not Found");
    }
}
```

6. Count Even and Odd Numbers

```
public class EvenOddCount {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 4, 5, 6};
        int even = 0, odd = 0;

        for (int num : arr) {
            if (num % 2 == 0)
                even++;
            else
                odd++;
        }

        System.out.println("Even = " + even);
        System.out.println("Odd = " + odd);
    }
}
```

7. Sort Array (Ascending)

```
public class SortArray {
    public static void main(String[] args) {
        int[] arr = {5, 3, 8, 1, 2};

        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] > arr[j]) {
```

```
                int temp = arr[i];
                arr[i] = arr[j];
                arr[j] = temp;
            }
        }
    }

    for (int num : arr) {
        System.out.print(num + " ");
    }
  }
}
```

8. Find Second Largest Element

```
public class SecondLargest {
  public static void main(String[] args) {
    int[] arr = {12, 45, 7, 89, 23};

    int largest = Integer.MIN_VALUE;
    int second = Integer.MIN_VALUE;

    for (int num : arr) {
      if (num > largest) {
        second = largest;
        largest = num;
      } else if (num > second && num != largest) {
```

```
            second = num;
        }
    }

    System.out.println("Second Largest = " + second);
  }
}
```

9. Copy One Array to Another

```
public class CopyArray {
  public static void main(String[] args) {
    int[] arr = {1, 2, 3, 4};
    int[] copy = new int[arr.length];

    for (int i = 0; i < arr.length; i++) {
      copy[i] = arr[i];
    }

    for (int num : copy) {
      System.out.print(num + " ");
    }
  }
}
```

10. Find Duplicate Elements

```
public class DuplicateElements {
    public static void main(String[] args) {
        int[] arr = {1, 2, 3, 2, 4, 1};

        for (int i = 0; i < arr.length; i++) {
            for (int j = i + 1; j < arr.length; j++) {
                if (arr[i] == arr[j]) {
                    System.out.println("Duplicate: " + arr[i]);
                }
            }
        }
    }
}
```